WORKING REFERENCE · LOOP TUNING

Overnight Basal Plan

Andrew · Omnipod + Loop · Dexcom G7 · dorm since Aug 21 · **data through Aug 23, 2026**

> Personal working notes from Nightscout data. One change at a time; watch, then decide. Not medical advice.

WHERE THINGS STAND

Overall control	mean ~110, GMI ~5.9%
Time below 54 (last 2 wks)	1.8 – 3.1%
Recent overnight nadirs	40 · 43 · 50
Bolus insulin	33.9 U/day (was 43.3)
Scheduled basal	38.05 U/day
Basal Loop actually gives	~24.5 U/day

Average glucose is fine — this is a **hypoglycemia exposure** problem, not a control problem. Two things changed in August: marching band added activity, and he moved to a dorm where he eats late.

The schedule has drifted **56% above what Loop actually delivers**. Loop suspends 27–38% of the overnight hours holding it down — and the scheduled rate is what the pod falls back to if Loop stops.

THE CHANGE · OVERNIGHT ONLY

```
22:30  2.20 → 1.70
00:00  1.45 → 1.20
02:00  1.30 → 1.00
04:30  1.35 → unchanged
09:30+ daytime unchanged
```

Daily total	38.05 → **36.05 U** (−5.3%)
All of it between	22:30 and 04:30
If it under-delivers	1.55 / 1.10 / 0.90

HOUR	SCHED	LOOP GIVES	SUSP.
22:00 – 23:00	2.20 – 2.40	1.28 – 1.32	36–38%
00:00 – 01:00	1.45	0.88 – 0.92	31–33%
02:00 – 03:00	1.30	0.72 – 0.90	27–37%
04:00 – 08:00	1.30 – 1.35	1.00 – 1.11	10–16%

The schedule sits **56% above what Loop actually delivers**, and Loop suspends a third of the night holding it down. **04:00–08:00 is the exception** — Loop runs near the programmed rate there with no lows, so that block stays as it is.

This lowers real delivery, not just the fallback: during the ~65% of the night Loop isn't suspended, a lower baseline means less insulin. It doesn't cap Loop — it can still temp above schedule.

ALONGSIDE IT · LATE CORRECTIONS

Manual boluses add **3.74 U/night between 22:00 and 02:00**, at a median BG of 118–150 — dosing ahead of a rise rather than treating a high. That insulin is still working at 3–4am.

- ✕ **No manual correction under ~150 after 10pm.** 60% of them go in below 150. Late-meal peaks are modest anyway — median 159, only 3 of 12 passed 180.
- ✕ **No Rage mode or fake carbs at night.** Both recruit subcutaneous insulin whose 6-hour tail lands at 3–4am.
- ✓ **Afrezza instead** if a late correction is genuinely needed — ~90-minute action, gone before 2am. Substitute for the subcutaneous dose, don't add to it.

On Aug 22, a Rage correction at 01:00 with BG **127** was followed by **43** ninety minutes later.

WHAT TO WATCH

Run: `python3 check_response.py` — it already excludes sensor warmup and flags G7 compression.

Working: overnight (00:00–05:00) time below 54 drops toward <1%, no nadir under 60, and morning (07:00–09:00) mean stays under about 140.

Too far: morning mean climbs past ~160, or he starts correcting in the morning. Back the overnight rates up halfway.

Not enough: severe lows persist at 00:00–05:00 after 5–7 clean nights → go to the second step (1.55 / 1.10 / 0.90).

Ignore: single bad nights, and anything in the first 24h of a new sensor. Judge on 5–7 nights, excluding warmup.

NOT CHANGING

SETTING	CURRENT	WHY NOT
Correction target	100 flat, 24h	Not changing targets by preference. Note it does mean Loop keeps correcting toward 100 overnight — 81% of its automatic overnight insulin goes in below 150.
ISF	53	The other lever on Loop's aggressiveness (53 → 58 would soften every correction). Unvalidatable while Afrezza is unlogged — hold.
Carb ratio	6	Late-meal dosing already runs slightly weaker than this (implied 6.6) and late-meal peaks are modest. Not the problem.
04:30–09:30 basal	1.35	The one block Loop accepts. Produces no lows. Leave it.

WHAT THIS DATA CAN'T SEE

Afrezza is invisible. Not logged, so every insulin total here is a floor and any correction analysis undercounts.

Fake carbs inflate carb totals. Small entries used to prompt Loop to correct aren't food.

Prebolus timing is invisible — carbs and insulin are logged at the same timestamp, so the log shows dose time, not eat time.

The activity sample isn't the semester. Early-start days (n=7) ran 4.0% below 54 vs 0.9% on late-start days (n=52) — but those were band camp plus one Saturday event, not a normal ~9am-class week. Anything activity-shaped needs re-deriving in a few weeks. **The scheduled-vs-delivered gap does not depend on this** — it holds on every day type.

G7 compression lows are frequent overnight — brief untreated dips that rebound on their own. Excluded from the numbers here; they would otherwise roughly double the apparent low rate.

Source: personal Nightscout instance. CGM values are 5-minute means, duplicate devices de-duplicated. Delivered basal reconstructed from temp-basal records; bolus attribution uses Loop's own `automatic` flag. Prepared 2026-08-23.